

Association between LifeStyle Factors and Diabetes Prevalence across countries

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Background

- Diabetes is one of the fastest-growing chronic diseases.
- Lifestyle factors (Diet Quality and Physical Activity) are known contributors to diabetes risk.
- Previous studies have primarily focused on individual risk factors rather than evaluating multiple determinants simultaneously across countries.

Objectives

- This study aims to examine the association between lifestyle factors and diabetes prevalence.
- Evaluate whether these two factors remain significant after controlling for nominal GDP per capita, healthcare expenditure, and median age.

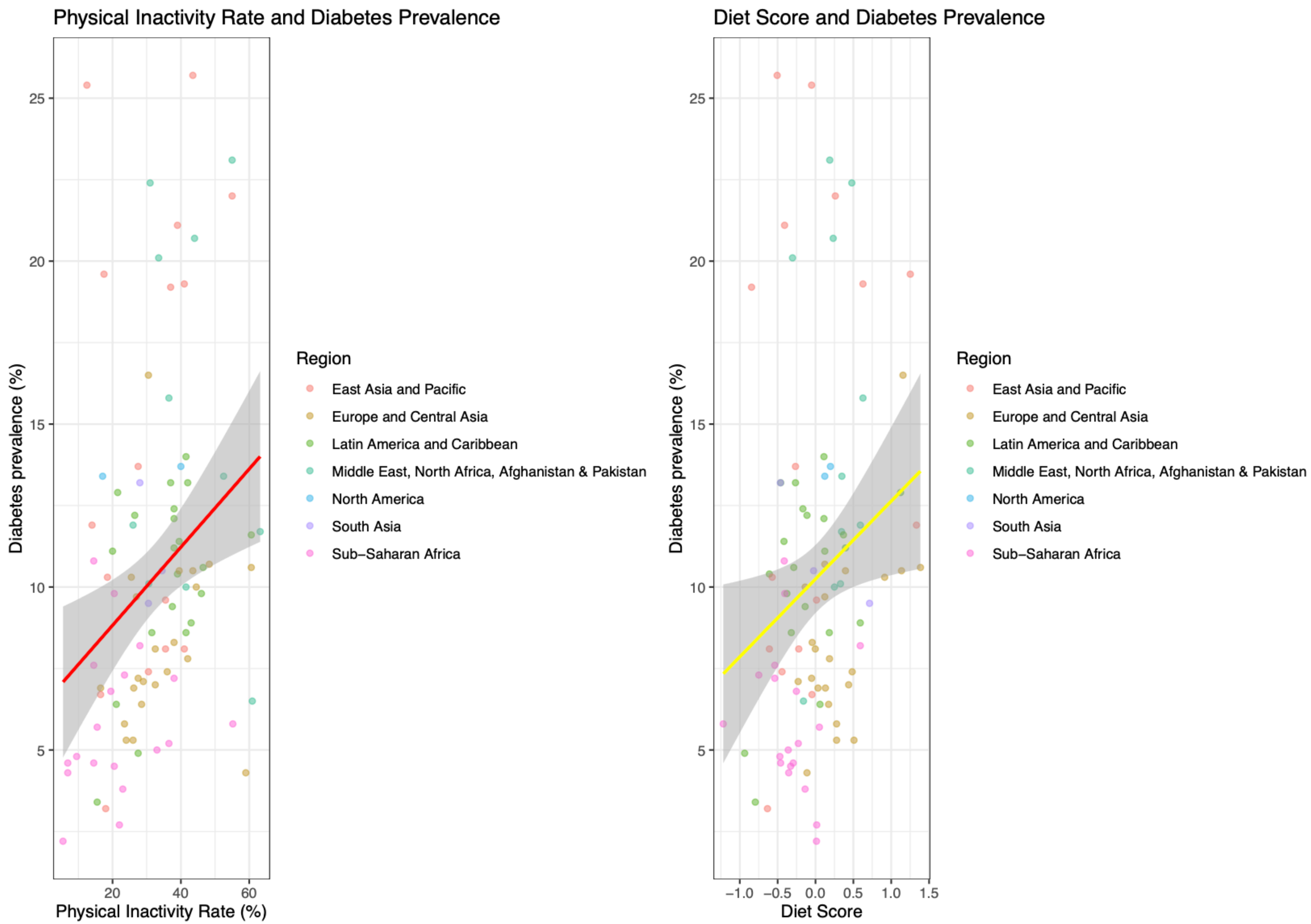
Method

- The study included data from 100 countries.
- Data were obtained from publicly available international databases, including WHO, the World Bank, NIH, World Population Review, and other publicly accessible sources.
- Outcome variable: Diabetes Prevalence
- Independent variables included diet quality score (z-score), physical inactivity rate, GDP per capita, healthcare expenditure, and median age.
- The diet score was constructed by standardizing per capita consumption of fruit, vegetables, wheat, and reverse-coding sugar consumption into z-scores.

Diet Score =
$$\frac{z(\text{Fruit}) + z(\text{Vegetables}) + z(\text{Wheat}) - z(\text{Sugar})}{4}$$

- Missing data were handled using multiple imputation (m=10).
- Ordinary Least Squares was estimated with a logit-transformed outcome variable.
- A pooled multiple linear regression model was fitted.
- Model assumptions (normality, homoscedasticity, and multicollinearity) were evaluated and found to be satisfied before interpreting the results.

Result



Pooled Multiple Linear Regression Results

logit-transformed diabetes prevalence; estimates pooled across 10 multiple imputed datasets

Predictor	Coefficients	95% Confident Interval		p-value
		Lower	Upper	
Intercept	−2.949	−3.467	−2.431	<0.001
GDP per capita	−0.021	−0.062	0.021	0.326
Healthcare expenditure	0.011	−0.028	0.050	0.575
Physical inactivity rate	1.332	0.422	2.242	0.005
Median age	0.006	−0.011	0.023	0.458
Diet quality score	0.283	0.054	0.512	0.016

Only Physical Inactivity Rate and Diet Score remained statistically significant after adjustment.

Discussion

- Physical Inactivity Rate remained the strongest predictor for diabetes prevalence after controlling for other factors ($p = 0.005$).
- A one-unit increase in physical inactivity rate was associated with a 1.332-unit increase in the logit-transformed diabetes prevalence.
- Diet quality score is significantly associated with diabetes rate ($p = 0.016$), emphasizing that overall dietary pattern may be more informative than individual food components.
- A one-unit increase in diet score was associated with a 0.283-unit increase in the logit-transformed diabetes prevalence.
- GDP per capita, healthcare expenditure, and median age don't have a significant relationship with the outcome variable.
- The positive association between diet quality score and diabetes prevalence should be interpreted with caution. This association should not be interpreted as indicating that healthier diets increase diabetes risk among individuals.

Limitations

- Some independent variables were collected from different reference years (2022 - 2024) due to data availability
- Some missing value required multiple imputation.
- The Diet Score was based on only available dietary variables and did not include several unhealthy dietary components due to data unavailability.

Conclusions

- Physical activity and diet quality were significantly associated with diabetes prevalence.
- Physical inactivity appears to be an important modifiable factor associated with diabetes prevalence across countries.
- Compared to demographic and economic factors, lifestyle factors appeared more important in analysing the diabetes rate.

References

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